

Lab 3 Lecture

Resistor Specifications

ECE-UY 2004 Fundamentals of Electric Circuits

What you need to know before going to the lab



Ideal Versus Real Resistors

Ideal



Practical



Specifications

- Tolerance
- Power Rating (maximum)
- Power derating
- Package: through hole, surface mount
- Composition: carbon, film, wirewound
- Cost



Resistor Value and Tolerance



Resistor Specifications: Tolerance

CF Series Carbon Film Resistor

General Purpose • Flame Retardant Coating • RoHS Compliant

Stackpole Electronics, Inc.

Resistive Product Solutions

EDUCATIONAL DATASHEET (Simplified)

For instructional use only

1.0 Electrical Specifications

Parameter	Specification
Nominal Resistance (R_N)	1,000 Ω (1 k Ω)
Tolerance	$\pm 5\%$
Resistance Range (5% Tol.)	950 Ω to 1.05 k Ω
Power Rating (at 70°C)	0.5 W
Maximum Working Voltage ⁽¹⁾	350 V
Maximum Overload Voltage	700 V
Dielectric Withstanding Voltage	500 V
Temperature Coefficient (TCR)	± 400 ppm/ $^{\circ}\text{C}$ (< 10 Ω) 0 to -400 ppm/ $^{\circ}\text{C}$ (10 Ω to 9.99 k Ω) 0 to -500 ppm/ $^{\circ}\text{C}$ (10 k Ω to 99 k Ω) 0 to -850 ppm/ $^{\circ}\text{C}$ (100 k Ω to 999 k Ω) 0 to -1500 ppm/ $^{\circ}\text{C}$ (1 M Ω and above)
Operating Temperature Range	-55 $^{\circ}\text{C}$ to $+155$ $^{\circ}\text{C}$

Nominal

Tolerance

1 k Ω with 5% tolerance

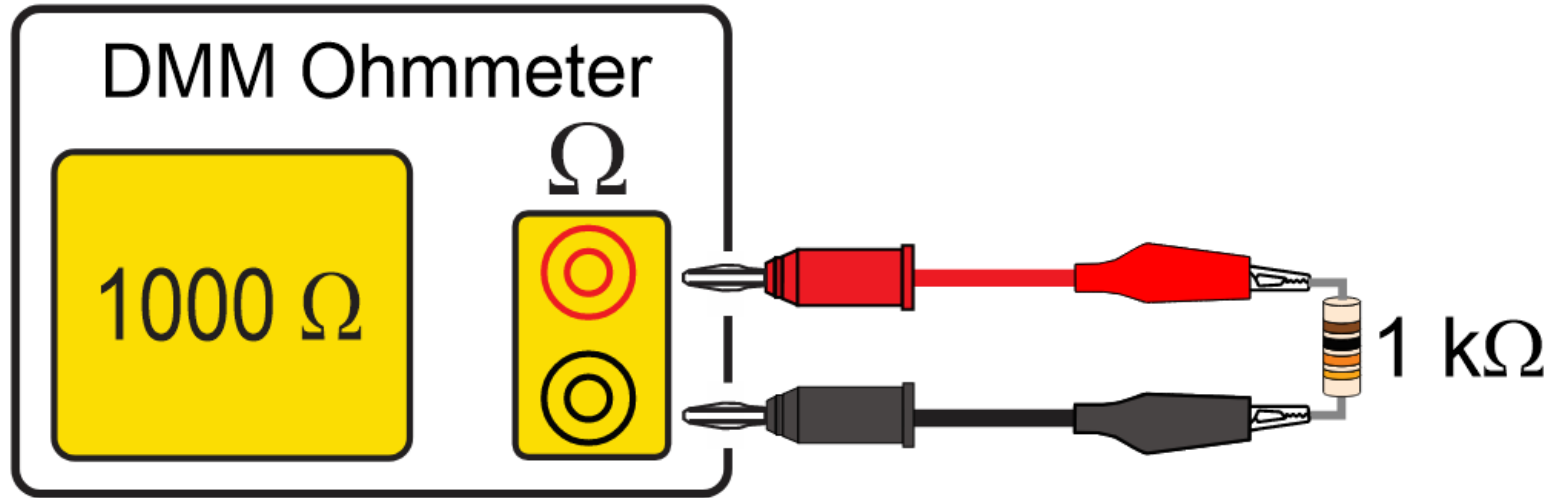
$1\text{k}\Omega \pm 0.05 \cdot (1\text{k}\Omega)$

$1\text{k}\Omega \pm 50 \Omega$

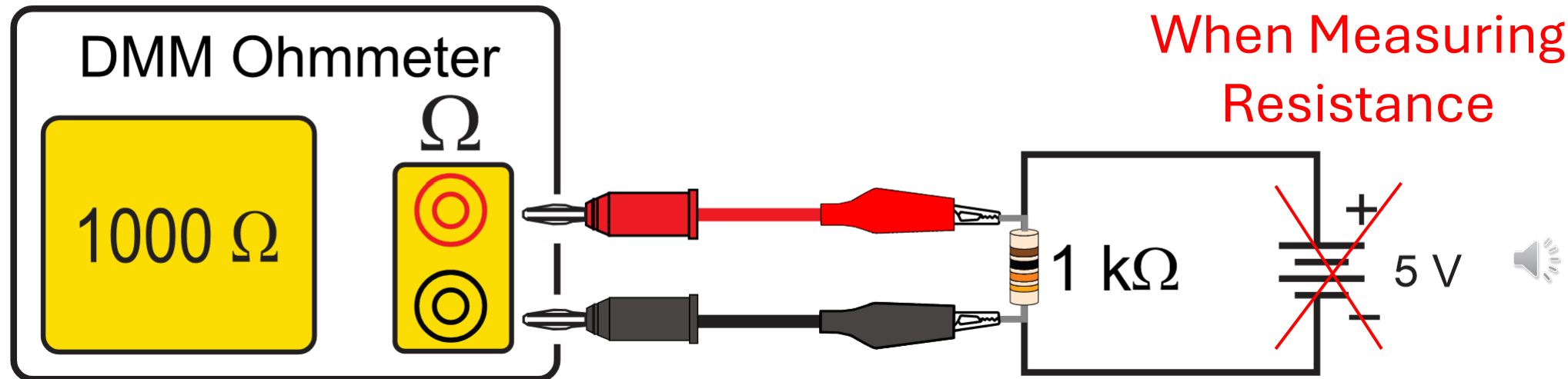
$950 \Omega \leq \text{range} \leq 1.05 \text{ k}\Omega$



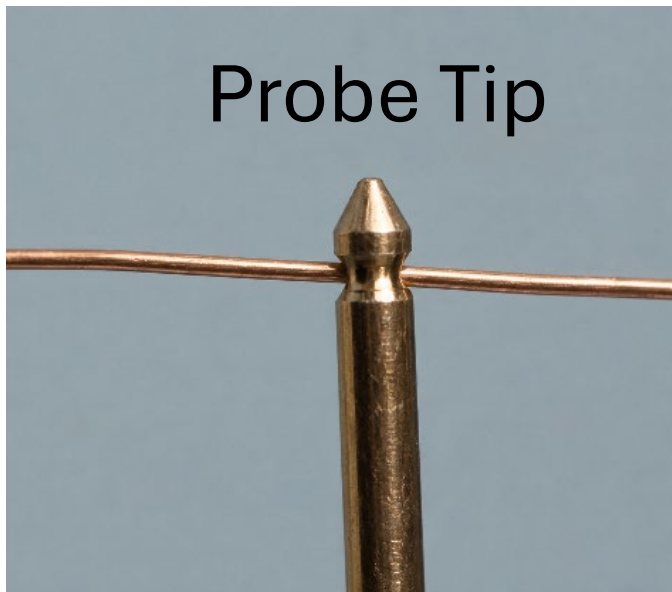
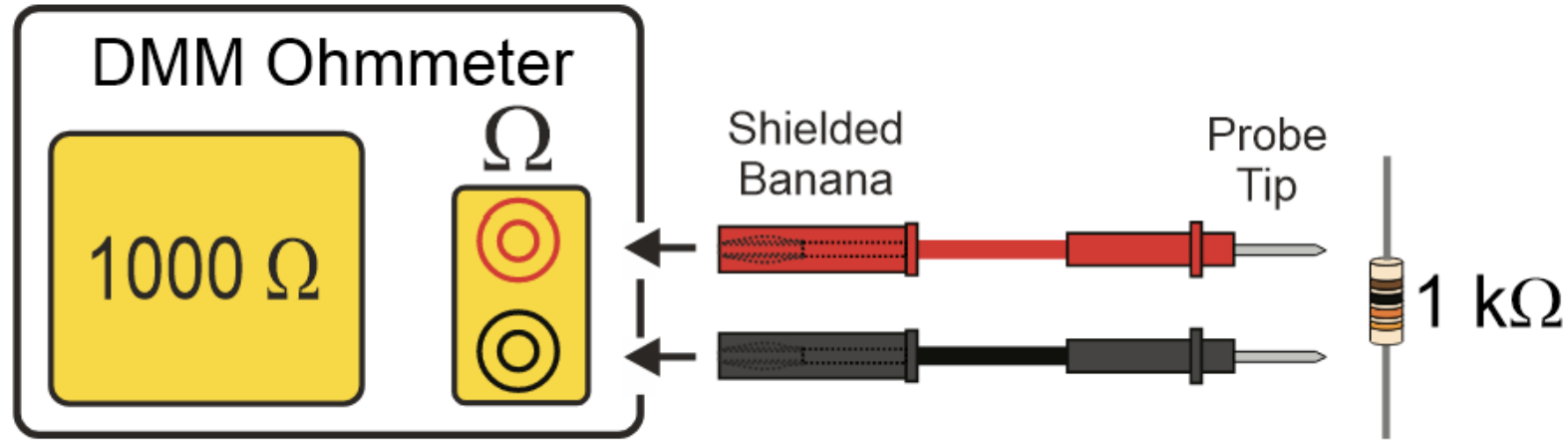
Measuring Resistance using the DMM



Do Not Power the Circuit
When Measuring
Resistance



Measuring Resistors using a Notched Test Lead



Probe Tip

Notched Tip

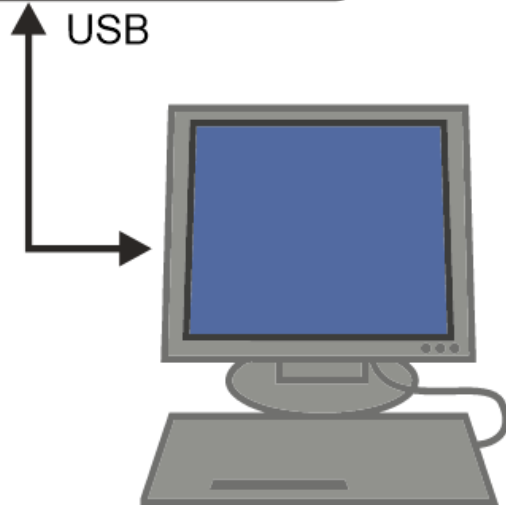
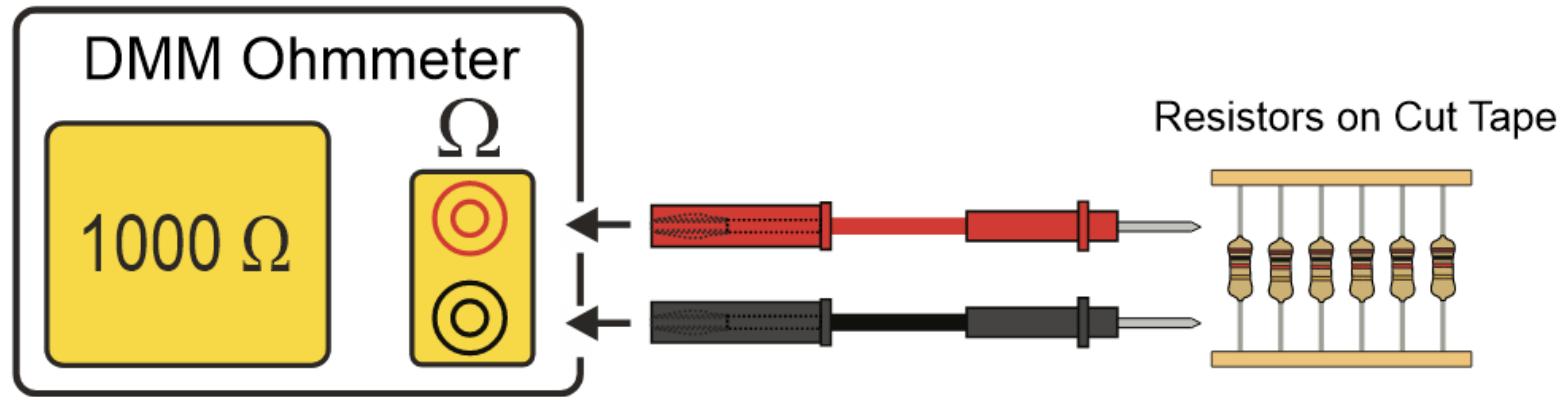
- Holds a round resistor lead in place
- Makes repeated measurements faster
- Maintain consistent electrical contact

Alligator Tip

- Hands free
- Reduce movement of the test lead



Measuring a Population of Resistors



5% Data	1% Data
1011	996
987	1017
994	1000
981	1003
982	1003
988	1009
984	1011
985	995
981	998
983	999

Measure 30 resistors

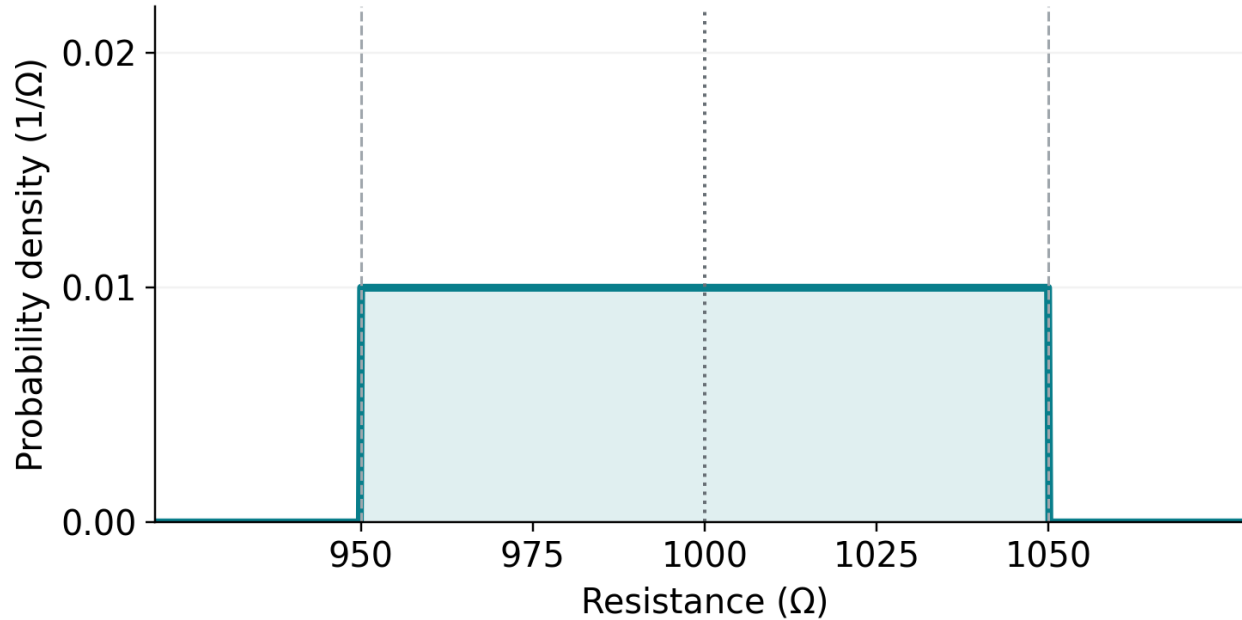
- 5%
- 1%

Resistors supplied on a reel



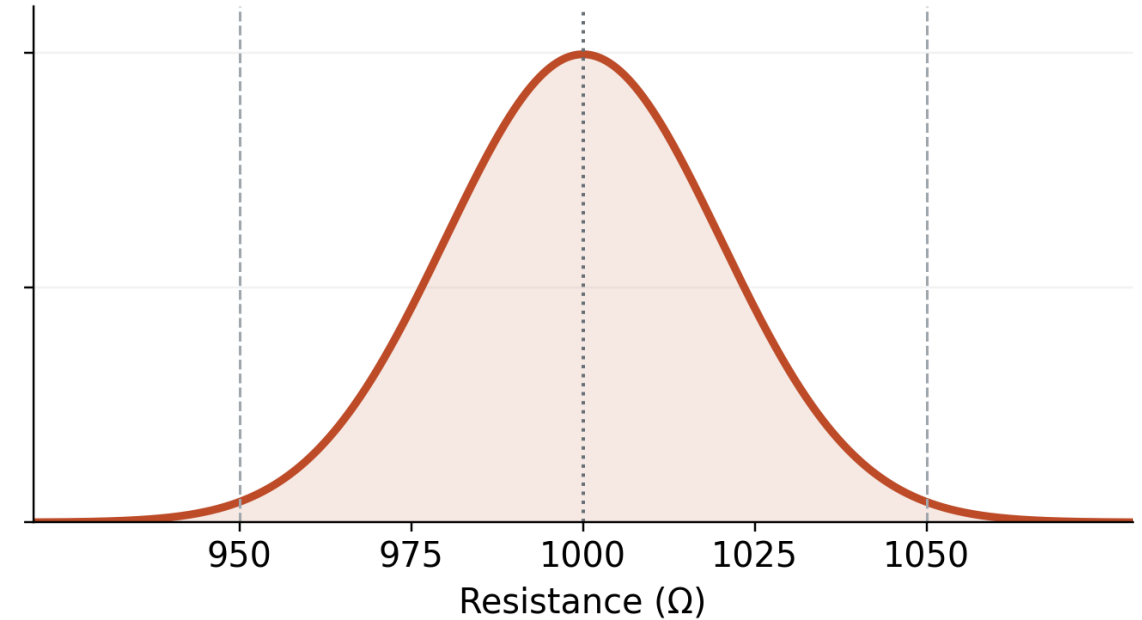
How Will Resistor Values be Distributed?

Uniform model



Equal density at every resistance from 950 to 1050 Ω .

Gaussian model

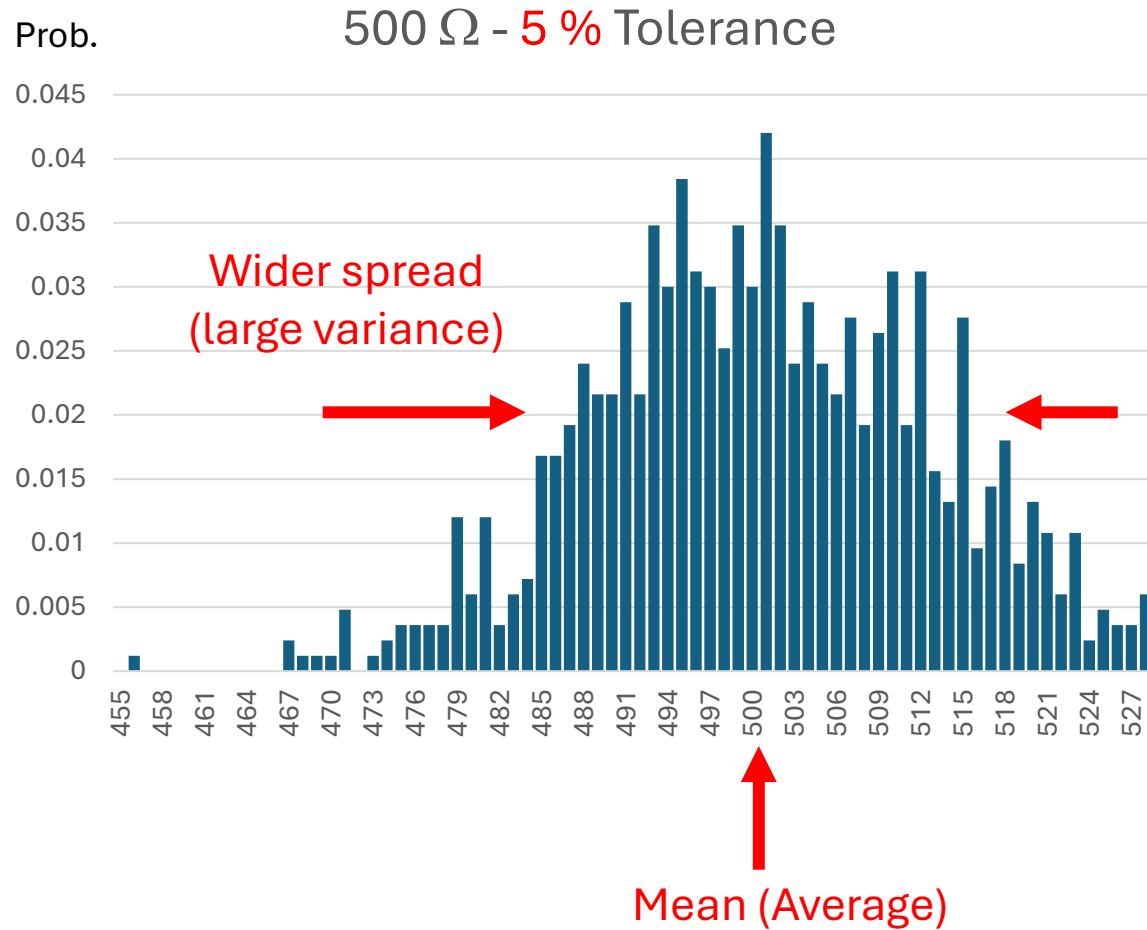


Values near 1000 Ω are more likely; density falls toward the edges.

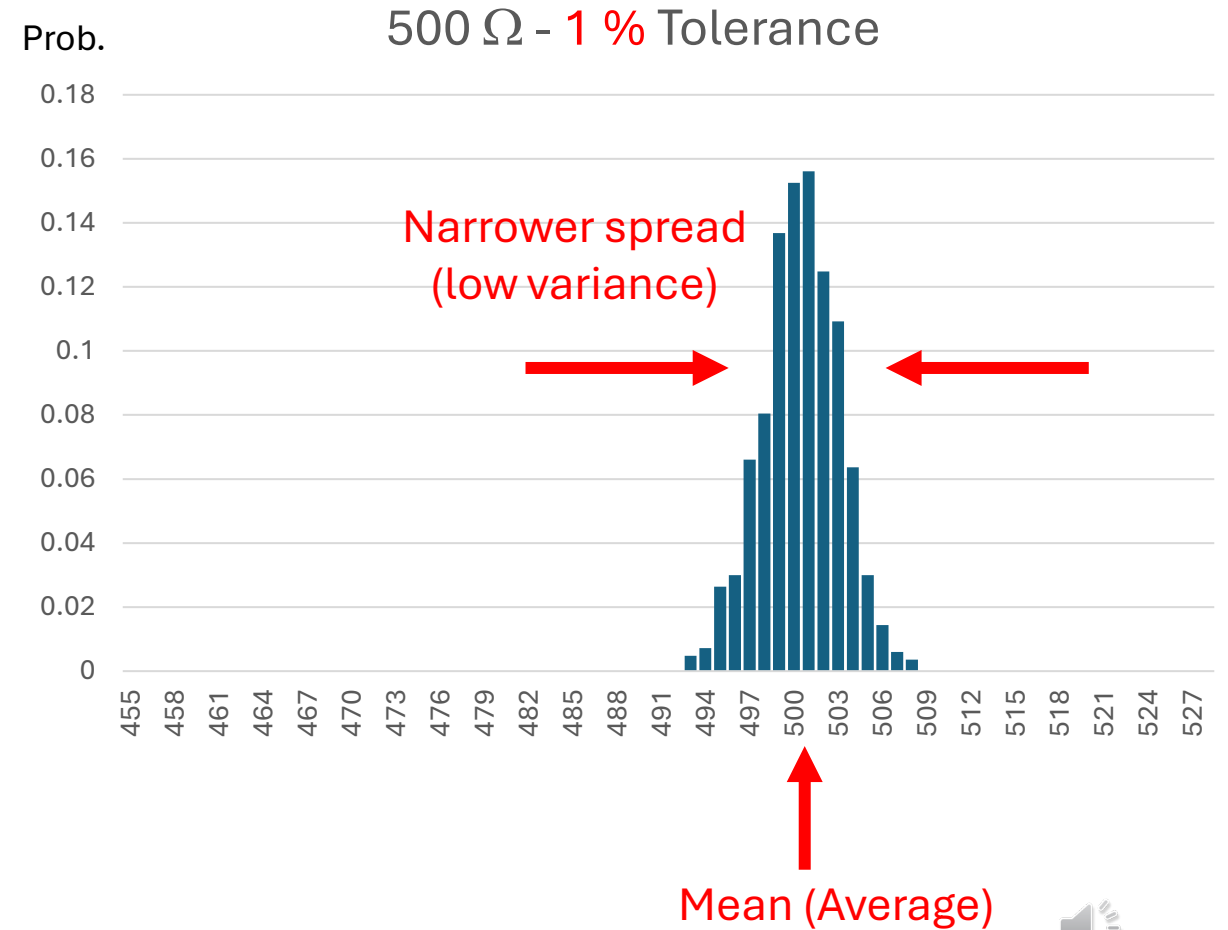


Real resistors tend to follow a Gaussian (bell-shaped) Curve

Example of 500 Ω resistor



Tolerance specifies the allowed range; the histogram shows how the measured values are distributed within it



Measured data from 500 resistors



Resistor Power Rating



Resistor Specifications: Power Rating

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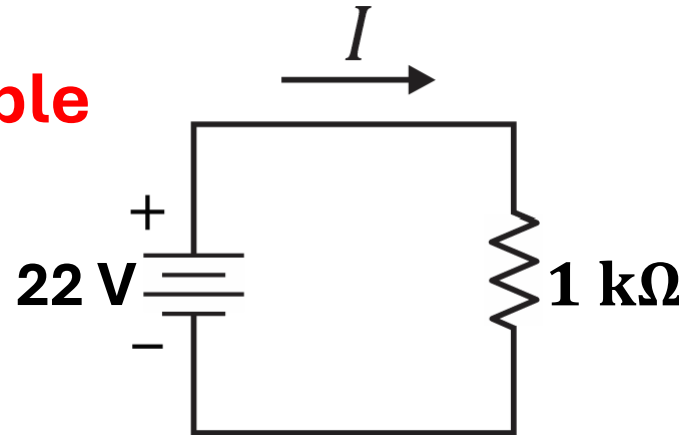
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Operating Temperature Range	-55 °C to +155 °C

Example



$$P = \frac{(22 \text{ V})^2}{1 \text{ k}\Omega} = 0.484 \text{ W}$$

Choose resistor with $P_{\text{rating}} > 0.484 \text{ W}$



Resistor Specifications: Power Derating

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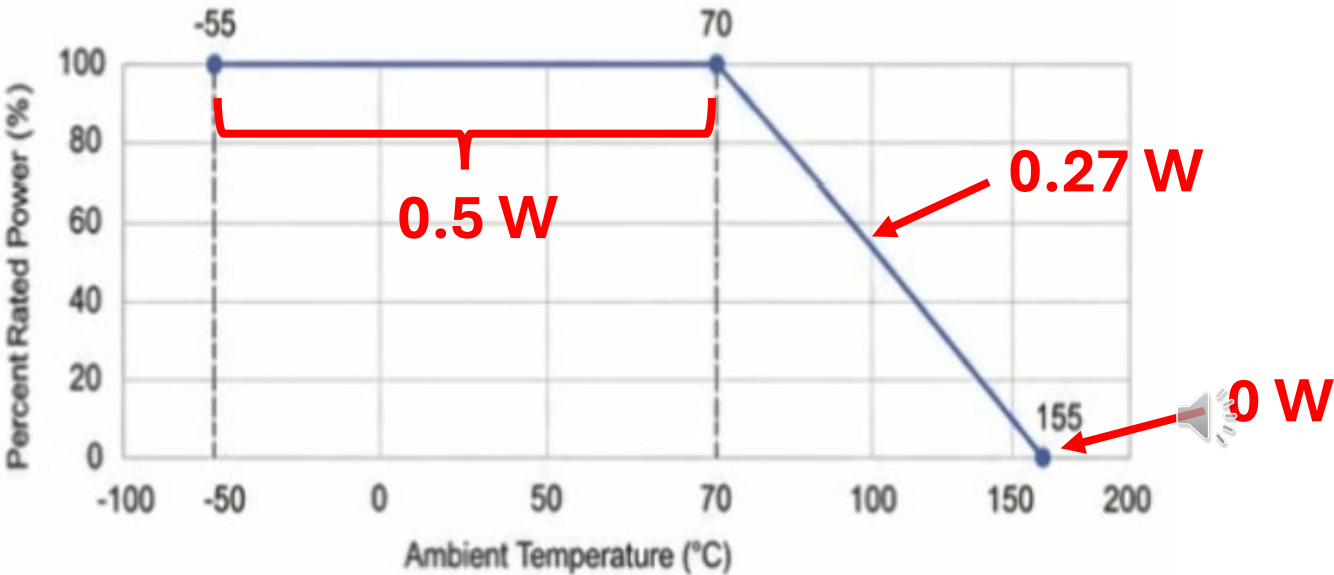
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3.0 Power Derating Curve

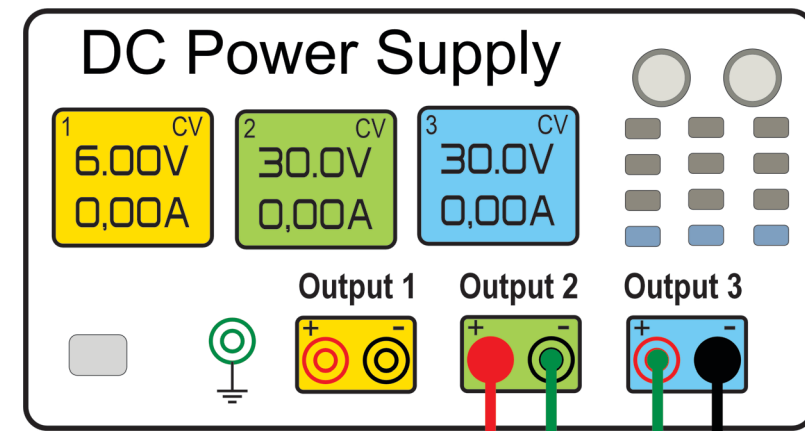
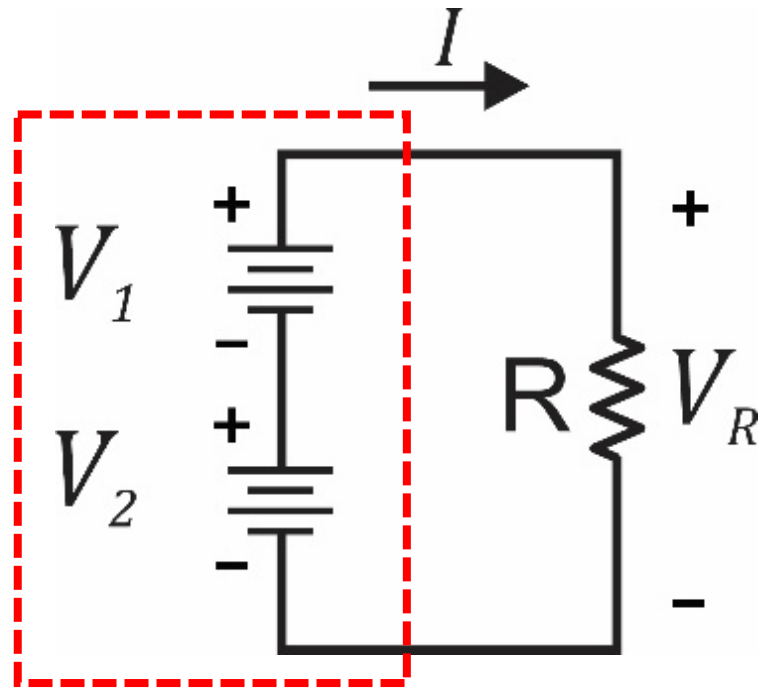
Resistors operated in ambient temperatures above 70°C must have their allowable power rating derated as shown.



Power Supply Voltage Settings



Connecting Power Supply Channels in Series



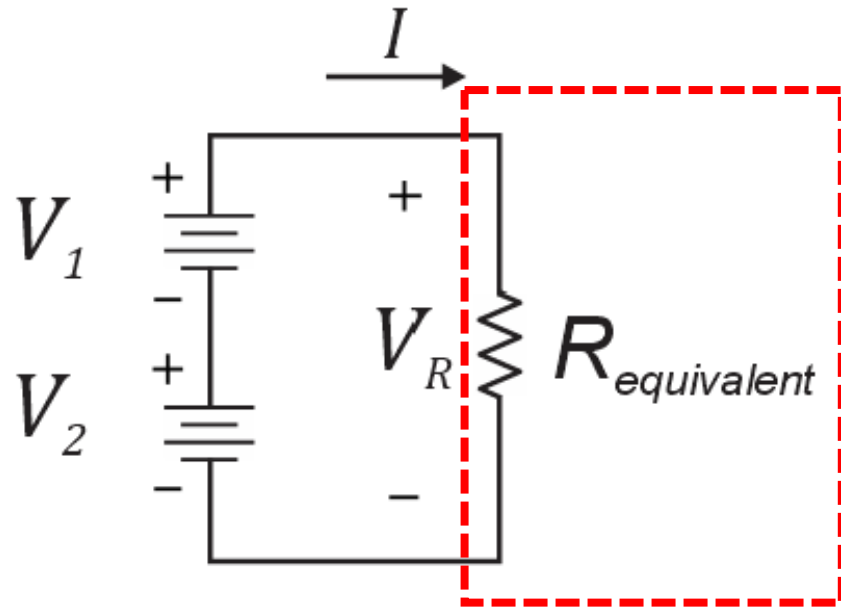
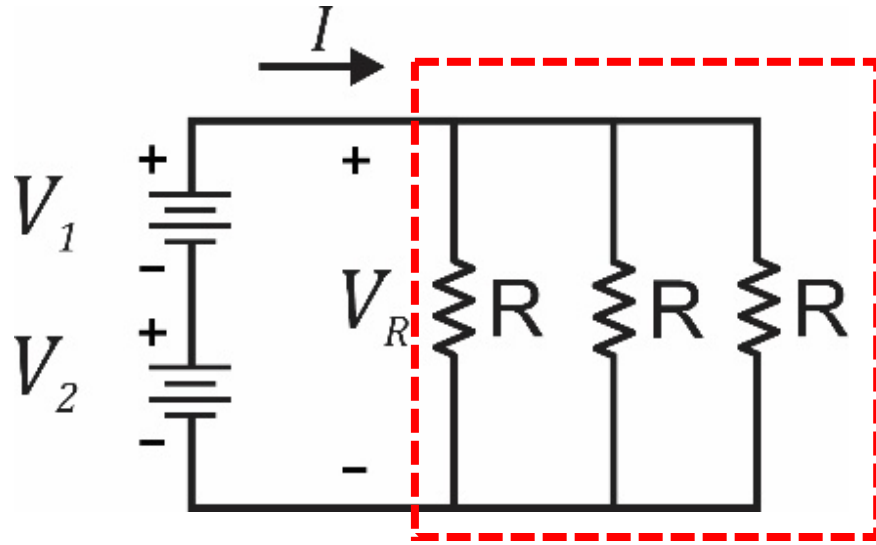
$+ V_R -$
Up to 60V



Parallel Resistors and Sharing the Power



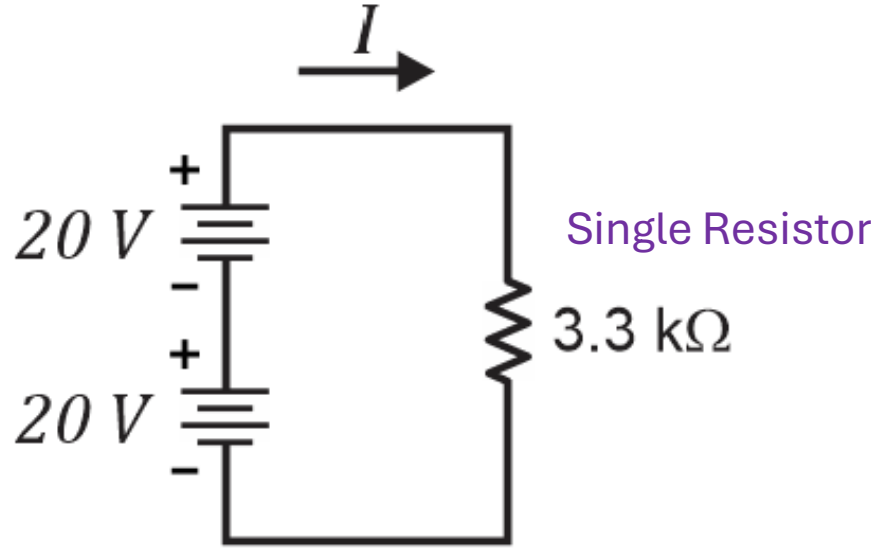
Parallel Resistors – Equivalent Resistance



$$\frac{1}{R_{equivalent}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

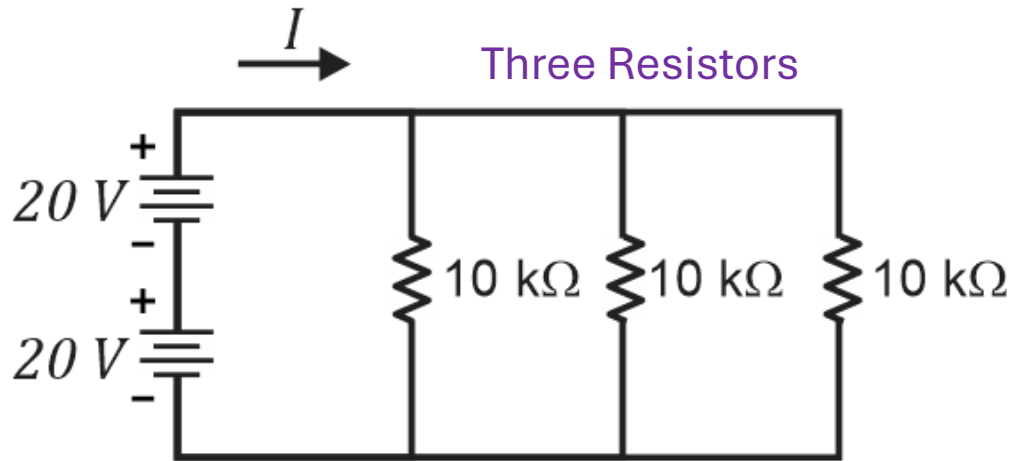


Example: Power Sharing



$$P = \frac{(40\text{ V})^2}{3.3\text{ k}\Omega} = 0.484\text{ W}$$

$3.3\text{ k}\Omega$ rated at 0.5 W



$$P = \frac{(40\text{ V})^2}{10\text{ k}\Omega} = 0.160\text{ W}$$

Each $10\text{ k}\Omega$ rated at 0.25 W

(note combined rating is now 0.75 W)

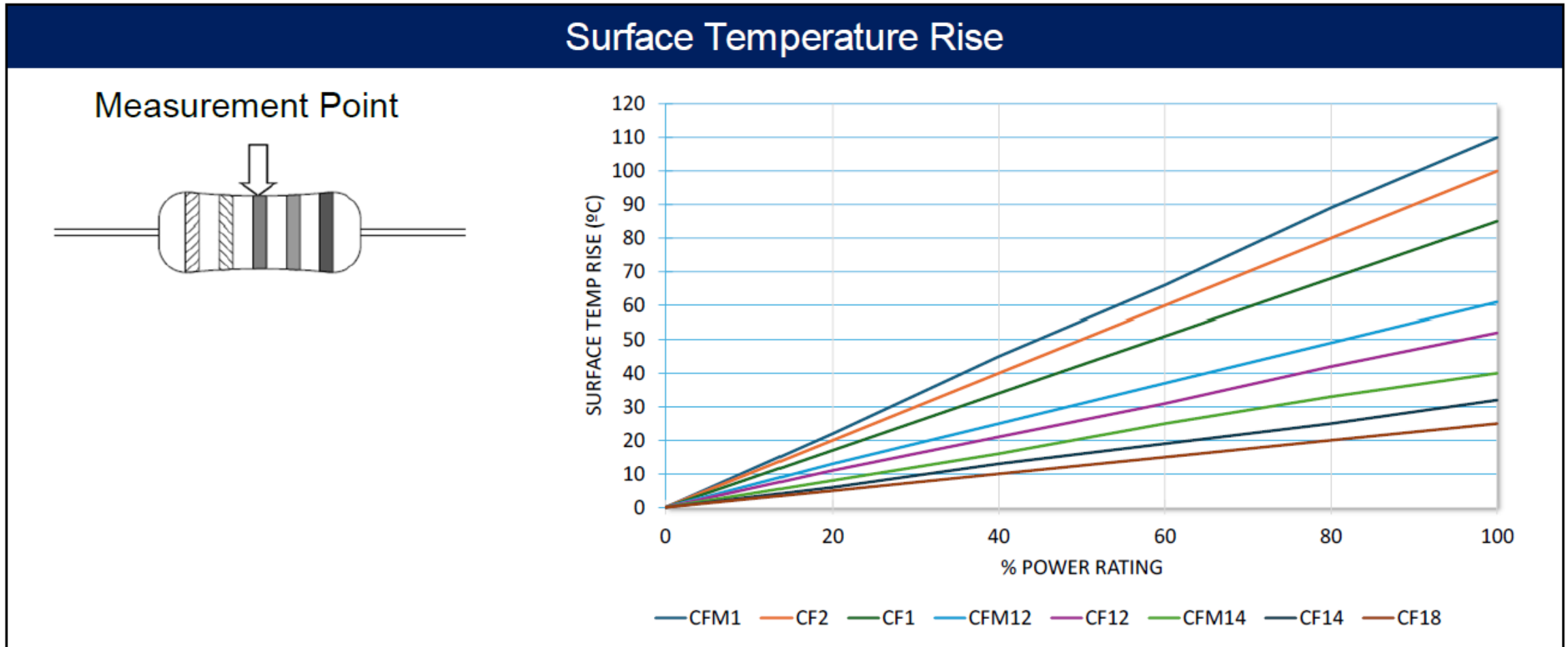
$$R_{\text{equivalent}} = 3.333\text{ k}$$



Appendix



Surface Temperature Power Derating



From stackpole digikey datasheet, using CF12 as an example

Environmental Temperature and Power Derating

Ambient temperature

The **ambient temperature** is simply the temperature of the surrounding air.

For example:

- Laboratory: **25°C**
- Inside a piece of equipment: **60°C**
- Near a hot power supply: **80°C**

The power derating curve assumes the resistor is operating in air at a given ambient temperature.

Surface temperature

The **surface temperature** is the actual temperature of the resistor body while it is dissipating power.

For example, in a lab:

- Ambient temperature = **23°C**
- Applied power = **0.5 W**
- Resistor surface temperature = **115°C**

Notice that the resistor is much hotter than the surrounding air because it is generating heat internally.

Temperature Rise and Power Dissipation

Over the limited power and temperature range in your experiment, the resistor's steady-state temperature rise may be approximately proportional to power:

$$\Delta T \approx P\theta$$

where ΔT is the temperature rise above room temperature and θ is the effective thermal resistance in °C/W.

Because

$$P = \frac{V^2}{R},$$

doubling the voltage produces approximately four times the power. If the resistor's mounting, airflow, probe position, and effective thermal resistance remain approximately unchanged, then the temperature rise can also be approximately four times larger.

For example:

- At 20 V: $P \approx 0.12$ W, temperature rise perhaps 10°C.
- At 40 V: $P \approx 0.48$ W, temperature rise perhaps 40°C.

That is a physically reasonable result. It would not be expected to remain exactly proportional over every temperature range because convection, radiation, resistance change, and heat conduction through the leads vary with temperature.